Graduates Hub

SEO Implementation Report

A complete technical and content-level audit of all search engine optimisation structures implemented across **graduateshub.co.za**

Technical SEO | On-Page SEO | Structured Data | Content SEO | E-E-A-T | Performance

DOMAIN	PAGES AUDITED	SCHEMA TYPES
graduateshub.co.za	**30+**	**11**

REPORT DATE

April 2026

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1 Technical SEO Files

Core crawlability, indexability, and site configuration

`app/robots.ts` — Robots Configuration

- **User-Agent:** `*` — all crawlers allowed
- **Allow:** `/` — full site crawl permitted
- **Disallow:** `/api/` — API routes blocked from indexing
- **Host:** `https://www.graduateshub.co.za` — www enforced as canonical host
- **Sitemap:** `https://www.graduateshub.co.za/sitemap.xml`

`app/sitemap.ts` — Dynamic XML Sitemap

Programmatically generated sitemap covering all content types:

Content Type	Priority	Change Frequency	OG Images
Homepage	1.0	weekly	Yes
Category hub pages	0.9	weekly	No
Free course guides	0.9	monthly	No
Blog listing	0.9	daily	No
CV Builder / Roadmaps / Interview Prep	0.9	monthly	No
Individual category pages	0.8	weekly	Yes
Career roadmap pages	0.8	monthly	No
Interview prep pages	0.8	monthly	No
Course detail pages	0.7	monthly	No
About / Contact	0.5	monthly	No
Legal pages	0.3	yearly	No

`app/manifest.ts` — Web App Manifest (PWA)

- **Name:** "Graduates Hub" | **Short Name:** "Graduates Hub"
- **Display:** `standalone` (full PWA capability)
- **Language:** `en-ZA` (South African English)
- **Category:** `['education']`

- **Theme:** `#2563eb` (blue) | **Background:** `#ffffff`
- **Icons:** Maskable icons at 192×192 and 512×512

`next.config.ts` — Redirects & Headers

301 Permanent Redirects

- `graduateshub.co.za/*` → `www.graduateshub.co.za/*` — non-www canonicalisation
- `/articles` → `/blog` — URL consolidation
- `/affiliate-disclosure` → `/disclosure` — URL cleanup

Security Headers

- `X-Frame-Options: SAMEORIGIN` — clickjacking protection
- `Strict-Transport-Security: max-age=2592000; includeSubDomains` — HTTPS enforced

Image Remote Patterns

- `images.unsplash.com`, `articles.graduateshub.co.za`, `*.gravatar.com`

`app/opengraph-image.tsx` & `app/twitter-image.tsx` — Dynamic OG Images

- **Dimensions:** 1200×630px (optimal social sharing ratio)
- **Format:** PNG, generated at request time via Next.js ImageResponse
- **Content:** Branded gradient background, logo, headline, feature badges
- **Alt:** "Graduates Hub - Free online courses with certificates"

2 Metadata Implementation

Title tags, descriptions, canonicals, and page-level overrides

`app/layout.tsx` — Root Metadata (Inherited by All Pages)

- **metadataBase:** `https://www.graduateshub.co.za`
- **Title template:** `"%s | Graduates Hub"`
- **Default title:** "Graduates Hub - Free Online Courses with Certificates"
- **Description:** "Empowering learners worldwide to achieve their goals with completely free, high-quality online education across IT, Business, Languages, and more."
- **openGraph.type:** website | **openGraph.siteName:** "Graduates Hub"
- **twitter.card:** summary_large_image
- **appleWebApp:** capable: true, statusBarStyle: "default"

Page-Level Metadata — All Routes

Page / Route	Title Strategy	Canonical	OG Image
`/` Homepage	Custom keyword-rich title	`/`	Generated
`/blog`	"Career Guides & Study Tips"	`/blog`	Default
`/blog/[slug]`	Dynamic from WordPress API	Per post	Featured image
`/course/[courseId]`	"{title} - Free Online Course"	Per course	Course image
`/category/[categoryId]`	"{Category} Free Courses (2026)"	Per category	Generated
`/career-roadmaps/[slug]`	"{title} (2026)"	Per roadmap	Default
`/interview-prep/[slug]`	"{role} CV & Interview Prep (2026)"	Per prep	Default
`/free-courses-with-certificates`	"Best Free Online Courses (2026)"	Static	Default
`/free-ai-courses-for-beginners`	"Best Free AI Courses (2026)"	Static	Default
`/free-digital-marketing-courses`	"Free Digital Marketing Courses"	Static	Default
`/free-courses-for-software-developers`	"Free Software Dev Courses"	Static	Default

Page / Route	Title Strategy	Canonical	OG Image
`/free-business-courses`	"Free Business Courses"	Static	Default
`/best-alison-courses-with-certificates`	"Best Alison Courses"	Static	Default
`/about`	"About Graduates Hub – Meet the Team"	`/about`	Default
`/search`	Static — **noindex**	—	—
`/privacy`, `/terms`, `/disclosure`	Legal page titles	Static	Default

generateMetadata — Dynamic Routes

The following routes use Next.js `generateMetadata()` for fully dynamic title/description/OG data:

- `app/blog/[slug]/page.tsx` — fetches from WordPress REST API
- `app/course/[courseId]/page.tsx` — fetches from course data source
- `app/category/[categoryId]/page.tsx` — per-category keyword-rich metadata
- `app/career-roadmaps/[slug]/page.tsx` — from roadmap data
- `app/interview-prep/[slug]/page.tsx` — from prep data

3 Structured Data / JSON-LD

Schema.org markup for rich results and knowledge graph signals

Schema Type	Implemented On	Key Fields
`Organization`	Homepage, About	name, url, logo, founder (sameAs: LinkedIn)
`WebSite` + `SearchAction`	Homepage	Enables Google Sitelinks Search Box in SERPs
`ItemList`	Homepage, Category pages, Guides	position, name, description, url per item
`Course` + `CourseInstance`	Every course detail page	name, description, provider, courseMode: Online, offers (price: 0 ZAR)
`Blog`	Blog listing page	name, url, description, publisher, first 6 BlogPost items
`BlogPosting` / `Article`	Every blog post	headline, datePublished, dateModified, image, author, publisher
`BreadcrumbList`	Blog, Courses, Roadmaps, Interview Prep, Guides, Contact	Full path from Home → current page, all links active
`HowTo`	Career roadmap pages	name, description, totalTime, step array (roadmap stages)
`FAQPage`	Homepage, all guide pages, category pages, interview prep	mainEntity array of Question/acceptedAnswer pairs
`Person`	About page	name, jobTitle, description, url, sameAs (LinkedIn), worksFor
`WebPage`	Free course guide pages	name, url, description, datePublished, dateModified

Notable Implementations

WebSite + SearchAction (Homepage)

Signals Google to show a Sitelinks Search Box in the SERP. Target URL:

`https://www.graduateshub.co.za/search?q={search_term_string}`

Course Schema with Free Offer

Every course page declares `offers.price: "0"` and `offers.priceCurrency: "ZAR"`, qualifying for Google's free course rich results in Search.

FAQPage Schema (Interactive FAQ Component)

The FAQ component embeds `<script type="application/ld+json">` inline, making each FAQ eligible for Google's FAQ rich results (expandable Q&A in SERPs). Used on: Homepage, every guide page, category pages, every interview prep page.

HowTo Schema on Career Roadmaps

Each roadmap stage maps to a `HowToStep`, including `totalTime` extracted from the "time to job-ready" field. Eligible for rich results in how-to queries.

Person Schema with sameAs (About Page)

Both founders (Jason Sadiki, Ndulamiso Mamburu) are described with full `Person` schema including LinkedIn and personal website `sameAs` links, contributing to Google's Knowledge Graph entity recognition.

4 Open Graph & Twitter Cards

Social sharing metadata for Facebook, LinkedIn, X/Twitter

Root Configuration (all pages inherit)

- `og:site_name` = "Graduates Hub"
- `og:type` = "website" (overridden to "article" on blog posts)
- `og:image` = graduates-hub-logo.png (1200×630 fallback)
- `twitter:card` = summary_large_image

Page-Level Overrides

Page Type	og:title	og:description	og:image Source
Homepage	Custom keyword title	Custom description	opengraph-image.tsx (generated)
Blog post	Post title from WP API	Post excerpt	WordPress featured image
Course page	"{title} - Free Online Course"	Course description	course.image
Category page	Category SEO title	Category description	opengraph-image.tsx (generated)
Career roadmap	"{title} (2026)"	Roadmap description	Root fallback
Interview prep	"{role} CV & Interview Prep"	Prep description	Root fallback
Guide pages	Guide title	Guide description	Root fallback
About	"About Graduates Hub – Meet the Team"	Founders description	Root fallback

Dynamic OG Image Generation

The homepage and category pages use `opengraph-image.tsx` — a Next.js `ImageResponse` component that renders a branded 1200×630 PNG on-demand. Content includes:

- Gradient background in brand blue
- Logo and site name
- Primary headline: "Free Online Courses with Certificates"
- Supporting subtitle with subject categories
- Feature badges: Self-paced, Certificates, Career-focused

Twitter uses an identical image via `twitter-image.tsx`.

5 Semantic HTML Structure

Heading hierarchy, landmark elements, and document outline

Heading Hierarchy

Level	Usage	Examples
`<h1>`	One per page — primary page topic	"Your free learning resource hub" (home), "{course.title}" (course), "{roadmap.title}" (roadmap)
`<h2>`	Major content sections	"Popular Guides", "Browse by Subject", "Why Choose Free Online Courses?", "The Team", "The Mission"
`<h3>`	Subsections, card titles, skill categories	"Technical Skills", "Soft Skills", individual skill items, course card titles

Semantic Landmark Elements

Element	Where Used	Notes
`<nav>`	Navbar, breadcrumb navigation	Breadcrumbs use `aria-label="Breadcrumb"`
`<main>`	Primary page content on all pages	flex-grow ensures it fills vertical space
`<article>`	Blog post content	Mirrors Article schema
`<section>`	FAQ blocks, guide sections, course sections	Groups related content
`<aside>`	Related articles, related guides sidebars	Supplementary to main content
`<footer>`	Site footer	50+ internal links across all sections

Breadcrumb Navigation

Visual breadcrumbs are implemented on all major content pages and match the `BreadcrumbList` schema exactly:

- Blog post: **Home › Blog › {Article Title}**
- Course: **Home › Categories › {Category} › {Course Title}**
- Career roadmap: **Home › Career Roadmaps › {Title}**
- Interview prep: **Home › Interview Prep › {Role}**
- Guides: **Home › Guides › {Guide Name}**

All breadcrumb items are functional links (not just text), and the wrapper uses `<nav aria-label="Breadcrumb">` for accessibility and crawler clarity.

6 Image Optimisation

next/image, alt text, responsive sizing, and lazy loading

next/image Usage

All images across the site use Next.js's `<Image>` component, providing automatic:

- WebP/AVIF format conversion
- Responsive srcset generation
- Lazy loading (below-fold default)
- Cumulative Layout Shift (CLS) prevention via explicit width/height

Responsive sizes Attributes

Component	sizes Value
CourseCard	`(max-width: 768px) 100vw, (max-width: 1024px) 50vw, 33vw`
Blog featured image	`(max-width: 768px) 100vw, 896px`
Related articles	`(max-width: 640px) 100vw, (max-width: 1024px) 50vw, 33vw`
Author profile photos	`(max-width: 768px) 128px, 256px`

Alt Text Strategy

Image Type	Alt Text Approach
Course card images	`alt={course.title}` — descriptive, keyword-relevant
Blog featured images	`alt={post.title.rendered}` — from WordPress API
Author photos (About)	"Jason Sadiki, Founder of Graduates Hub" / "Ndulamiso Mamburu, Co-Founder of Graduates Hub"
OG image	"Graduates Hub - Free online courses with certificates"
Decorative Lucide icons	`aria-hidden="true"` (excluded from accessibility tree)

Priority Loading

- `priority` prop set on blog featured images (above-the-fold, LCP element)
- `loading="lazy"` explicitly set on institution logos (below fold on homepage)
- All other images use Next.js default lazy loading

Remote Image Domains

- **images.unsplash.com** — fallback course imagery

- **articles.graduateshub.co.za** — WordPress blog featured images
- ***.gravatar.com** — author avatars (wildcard subdomain pattern)

7 Internal Linking Strategy

Navigation, footer, cross-content CTAs, and link equity distribution

Navigation Links (Header)

- **Explore dropdown:** All guide pages + all category pages
- **AI Tools dropdown:** AI-related resources
- **Header links:** Blog, About
- **Primary CTA:** "Find a Course" → `/search`

Footer Links (50+ Internal Links)

Footer Section	Link Count	Destination Types
Popular Guides	9	Free course guides, niche guides
Career Resources	8	Roadmaps, interview prep, CV builder
Categories	5	IT, Software Eng, Business, Accounting, Marketing
Legal & Company	7	About, Privacy, Terms, Disclosure, Curation Policy
Social / Community	3+	Facebook (189K+), LinkedIn, newsletter

Cross-Content CTAs

- Career roadmap pages → linked interview prep pages for that role
- Interview prep pages → linked course category
- Category pages → individual course pages + related guides
- Guide pages → relevant course categories + CV builder
- Blog posts → related articles sidebar + related guides sidebar

External Link Handling

- All external links: `target="_blank" rel="noopener noreferrer"`
- Affiliate links (Alison): include course-specific tracking, clearly labeled in ToS
- Author LinkedIn profiles: linked from About page and footer
- Facebook community: linked from footer and guide pages

Link Accessibility

- `aria-label` on course cards: `"View course details for {course.title}"`
- `aria-label="Toggle navigation menu"` on hamburger button

- `aria-expanded` + `aria-haspopup="true"` on dropdown triggers

8 Rendering & Performance

ISR, static generation, caching, and Core Web Vitals

Incremental Static Regeneration (ISR)

Page / Route	Revalidate	Strategy
Homepage (`/`)	300s (5 min)	ISR — frequent homepage content updates
Blog listing (`/blog`)	300s (5 min)	ISR — new posts published regularly
Blog post (`/blog/[slug]`)	3600s (1 hr)	ISR + dynamicParams: true
Course pages	None (static)	Fully static — course data is stable
Category pages	None (static)	Fully static — 5 known categories
Career roadmaps	None (static)	Fully static — pre-defined content
Interview prep	None (static)	Fully static — pre-defined content

Static Pre-rendering (generateStaticParams)

All dynamic routes are pre-rendered at build time:

- `app/blog/[slug]` — loops through paginated WordPress API to fetch all post slugs
- `app/course/[courseId]` — generates all course IDs from data
- `app/category/[categoryId]` — 5 categories pre-rendered
- `app/career-roadmaps/[slug]` — all roadmap slugs
- `app/interview-prep/[slug]` — all prep slugs

`dynamicParams: true` on blog posts ensures new posts published after the last build are still served via ISR (not a 404).

React cache() Deduplication

In `app/blog/[slug]/page.tsx`, the post fetch is wrapped in React's `cache()` function. This means `generateMetadata()` and the page component both call the same cached function — only one network request is made per render cycle, reducing latency and API load.

Streaming & Suspense

- `<Suspense>` used on the search page to stream search results progressively
- Provides a fallback loading state while data resolves, improving perceived performance

Monitoring

- **Vercel Speed Insights** — real-user Core Web Vitals (LCP, FID, CLS) tracking
- **Vercel Analytics** — traffic and page-level metrics
- Both injected in root layout, zero configuration required

9 E-E-A-T & Trust Signals

Experience, Expertise, Authoritativeness, Trustworthiness

AuthorByline Component — Expertise Signals

A dedicated `AuthorByline` component is displayed on guide pages, career roadmaps, and interview prep pages. It surfaces:

- Author full name and professional title
- Credentials and years of experience
- Links to LinkedIn profiles

This directly satisfies Google's E-E-A-T requirement that content be attributed to a demonstrably qualified person.

About Page — Founder Credentials

- **Jason Sadiki** — Technical SEO Specialist & Web Developer, LinkedIn linked
- **Ndulamiso Mamburu** — Accounting Science Graduate, employed at SARS, LinkedIn linked
- Both founders described via `Person` schema with `sameAs` links to LinkedIn and personal sites
- Organization schema links founders to the site entity

Trust & Transparency Pages

- **Curation Policy** (`/curation-policy`) — explains how courses are selected and vetted
- **Affiliate Disclosure** (`/disclosure`) — transparent about Alison affiliate relationships
- **Privacy Policy** (`/privacy`) — POPIA (South African GDPR equivalent) compliance implied
- **Terms of Service** (`/terms`) — defines user relationship

Social Proof

- Facebook community badge displaying **189,000+ followers** across guide and roadmap pages
- Course star ratings and review counts displayed in CourseCard components
- CPD (Continuing Professional Development) accreditation mentioned in category descriptions

Content Freshness Signals

- All guide pages and roadmap pages include "(2026)" in title tags — signals recency
- WebPage schema includes `datePublished` and `dateModified` on guide pages

- BlogPosting schema includes `datePublished` and `dateModified` on every post

10 Accessibility (SEO-Relevant)

ARIA attributes, keyboard navigation, and WCAG alignment

Element	Implementation	SEO Benefit
Course card links	`aria-label="View course details for {title}"`	Descriptive anchor text for crawlers
Navigation dropdowns	`aria-expanded`, `aria-haspopup="true"`	Crawlable interactive structure
Breadcrumb nav	`<nav aria-label="Breadcrumb">`	Crawler recognises breadcrumb pattern
Decorative icons	`aria-hidden="true"` on Lucide icons	Excludes decorative elements from accessibility tree
Interactive elements	`focus-visible:ring` focus states	Keyboard navigability (UX → lower bounce)
Mobile navigation	Hamburger with `aria-label="Toggle navigation menu"`	Accessible on all devices; mobile-friendly ranking signal

11 Analytics & Tracking

Measurement tools wired into root layout

Tool	ID / Config	Purpose
Google Analytics 4	`G-JF4KBF34HB`	Traffic, conversions, behaviour
Google AdSense	`ca-pub-7185526762692935`	Monetisation
Vercel Speed Insights	Root layout	Real-user Core Web Vitals
Vercel Analytics	Root layout	Page views, traffic sources
OneSignal	Push config	Push notifications / re-engagement

12 Summary Dashboard

At-a-glance coverage across all SEO dimensions

PAGES WITH METADATA

30+

Every route has explicit title, description, and canonical

SCHEMA TYPES DEPLOYED

11

Organization, Course, Blog, Article, HowTo, FAQ, Person, BreadcrumbList, ItemList, WebSite, WebPage

DYNAMIC ROUTES PRE-RENDERED

5

Blog, Course, Category, Roadmap, Interview Prep — all use generateStaticParams

INTERNAL FOOTER LINKS

50+

Distributing link equity across guides, categories, resources, and legal pages

301 REDIRECTS

3

Non-www → www, /articles → /blog, /affiliate-disclosure → /disclosure

STRUCTURED DATA PAGES

40+

FAQPage, BreadcrumbList, and Course schemas on the majority of indexable routes

Coverage by Dimension

SEO Dimension	Status	Notes
Technical files (robots, sitemap, manifest)	Complete	All core files present and configured
Redirects & canonicalisation	Complete	www enforced, legacy URLs redirected
Metadata (titles, descriptions, canonicals)	Complete	30+ pages, dynamic generation on all dynamic routes
Structured data / JSON-LD	Complete	11 schema types, all content verticals covered
Open Graph & Twitter Cards	Complete	Root config + per-page overrides + dynamic OG images
Semantic HTML & heading hierarchy	Complete	Single h1 per page, full landmark element usage

SEO Dimension	Status	Notes
Image optimisation	Complete	next/image, alt text, responsive sizes, priority flags
Internal linking	Complete	Nav, footer, cross-content CTAs, breadcrumbs
Rendering (ISR, SSG)	Complete	All dynamic routes statically generated or ISR
E-E-A-T signals	Complete	Author bylines, founder bios, trust pages, social proof
Accessibility (ARIA)	Complete	aria-labels, aria-expanded, focus states
Analytics & monitoring	Complete	GA4, Vercel Speed Insights, Vercel Analytics
Font preloading	Opportunity	No explicit font-display:swap or preconnect
Video schema	Opportunity	No video content currently; could add for tutorials
Hreflang / i18n	Opportunity	en-ZA only; expansion could target en-NG, en-KE